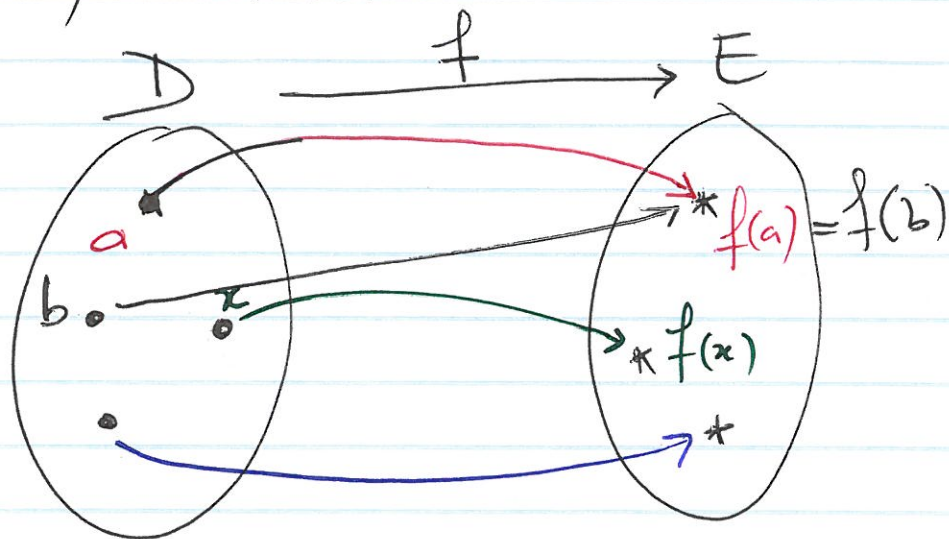


1.3 What is a function?

Definition: A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E .



Usually, we consider functions for which the sets D and E are sets of real numbers.

The set D is called the domain of f . The number $f(x)$ is the value of f at x and is read " f of x ". The range of f is the set of all possible values of $f(x)$.

as x varies throughout the domain.

A symbol that represents a number in the domain of f is called an independent variable. A symbol that represents a number in the range of f is called a dependent variable.

Example 1 : (a) $y = \sqrt{x} = f(x)$

x : independent variable

y : dependent variable

$$D_f = \{x / x \geq 0\} = [0, +\infty)$$

Since $\sqrt{x} \geq 0$ for all $x \in D_f$ then

$$R_f = [0, +\infty)$$

(b) $z = \sin(\theta) = h(\theta)$

θ : independent variable

z : dependent — " —

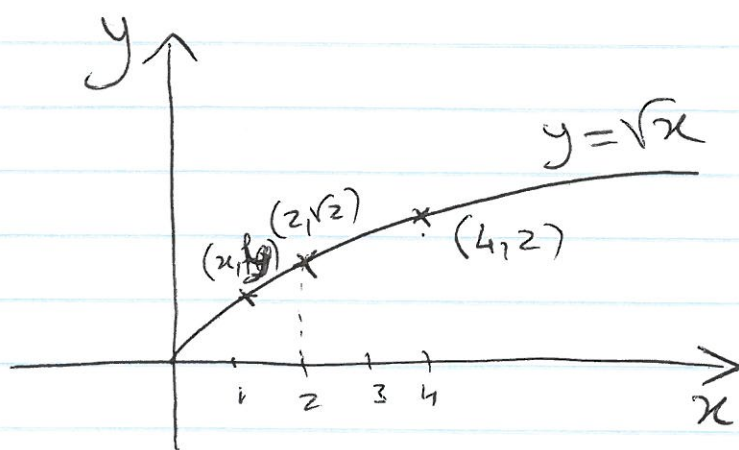
$$D_h = \mathbb{R} \quad ; \quad R_h = [-1, 1]$$

The graph of f is the set of ordered pairs $\{(x, f(x)) / x \in D\}$

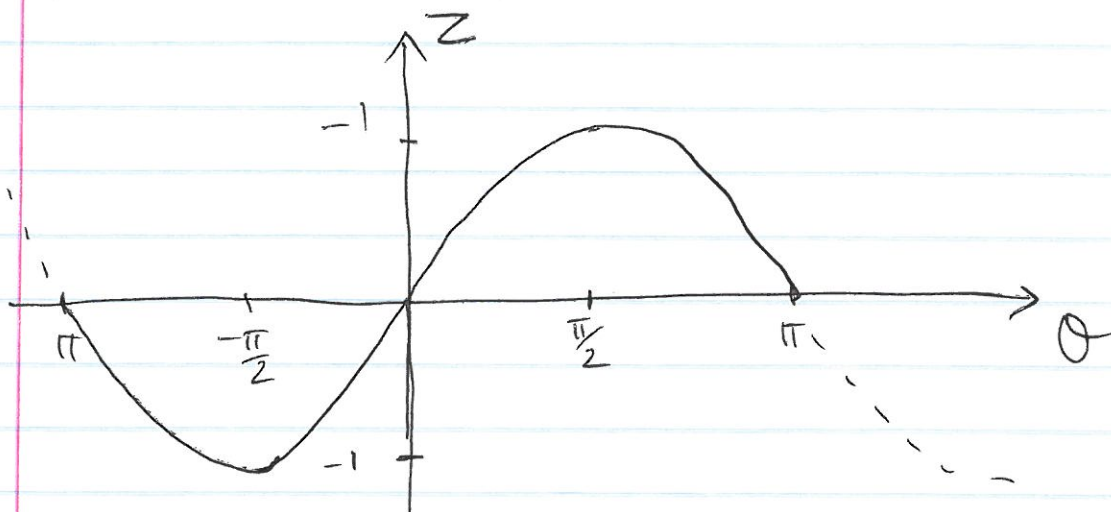
that is $\{(x, y) / x \in D \text{ and } y = f(x)\}$

Example 2 :

(a) $y = \sqrt{x}$



(b) $z = \sin(\theta)$



The four ways to represent a function:

- Verbally (by a description in words)
- numerically (by a table of values)
- visually (by a graph)
- algebraically (by an explicit formula)

Example 3:

(a) Paul is twice Jack's age. Together they are 60.

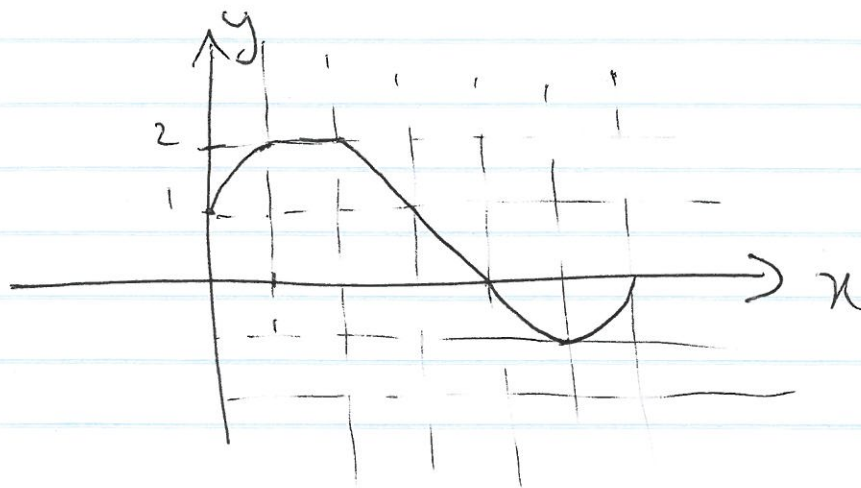
unknown
 $3x = 60$ is
an equation

Not a representation of
a function!!
..

(b)

t	Population (millions)
0	1650
10	1750
20	1860
⋮	⋮
⋮	⋮
⋮	⋮

(c)



$$(d) f(x) = x^2 + 1$$

Example 4 : Determine the domain of f , sketch its graph and find its range

$$(a) f(x) = 2x - 1$$

$$(e) f(x) = \frac{1}{x}$$

$$(b) f(x) = \sqrt{1-x}$$

$$(c) f(t) = -\frac{1}{t^2}$$

$$(d) f(u) = 3^u$$

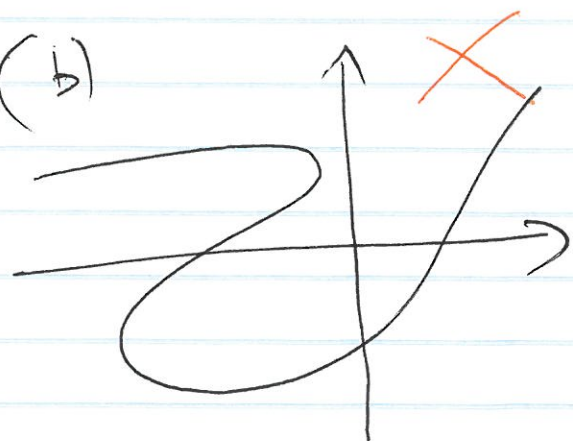
The vertical line test: A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

Example 5 :

(a)

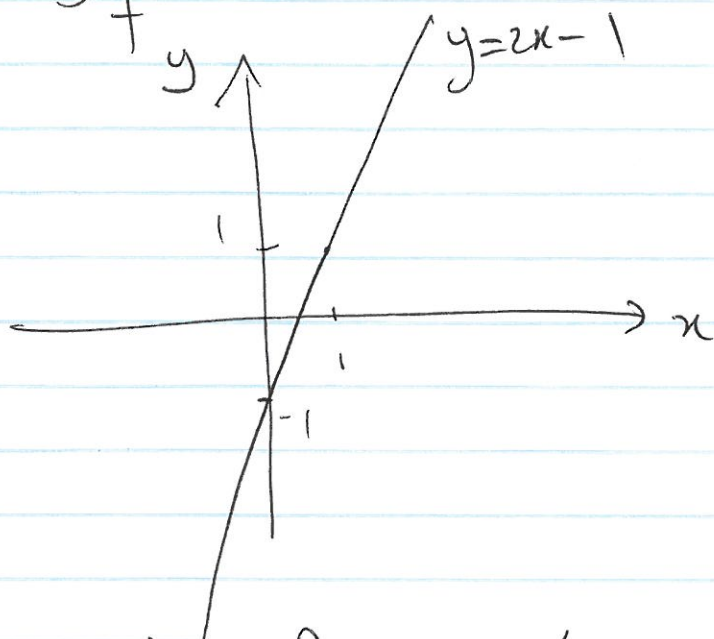


(b)



Example 4:

(a) $D_f = \mathbb{R}$

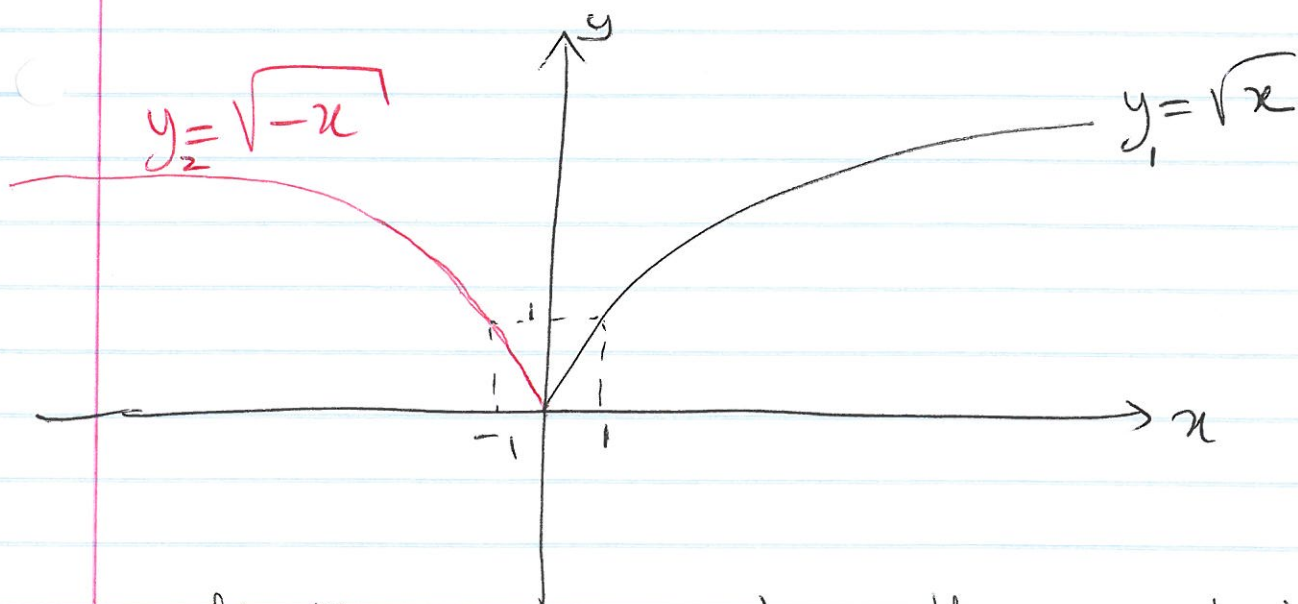


$R_f = \mathbb{R}$

(b) $D_f = \{x \in \mathbb{R} / 1 - x \geq 0\}$

$= \{x \in \mathbb{R} / x \leq 1\}$

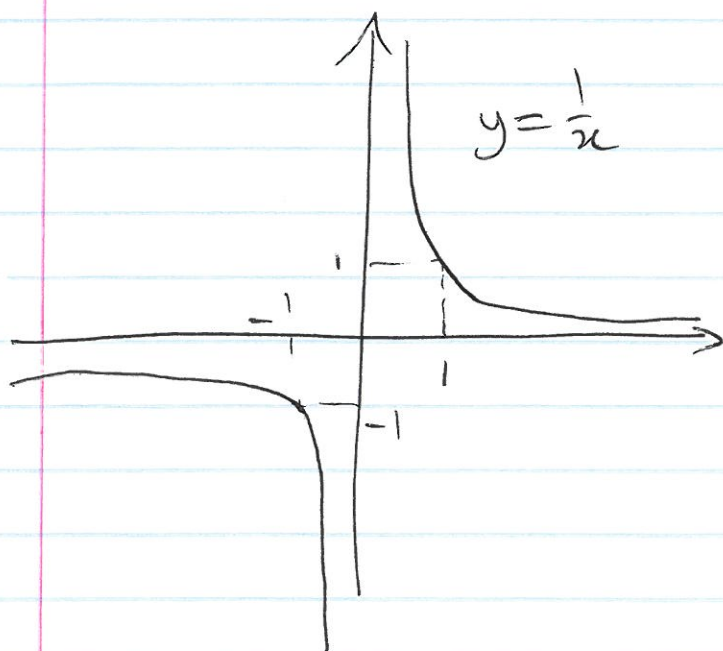
$= (-\infty, 1]$



y_2 reflection or symmetry with respect to the y-axis of y_1

$$R_f = [0, +\infty)$$

$$(e) \quad D_f = \{x / x \neq 0\} = (-\infty, 0) \cup (0, +\infty)$$



$$R_f = (-\infty, 0) \cup (0, +\infty)$$

Translation : "All x real numbers such that
 $x \geq 0$ "

$$D_f = \{x / x \geq 0\}$$